From "Truth Topology" to Readout Bottlenecks: Negative Replication on GSM8K and a Positive Representation–Decoder Dissociation in Procedural Micro-Worlds

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Abstract

This paper reports a multi-stage study of hidden-state geometry for correctness prediction in language-model reasoning. The original hypothesis was ambitious: perhaps global topological summaries of hidden-state traces could provide a robust signal of correctness or "truth." That hypothesis did not survive stronger checks. On GSM8K-style reasoning traces [4], dynamic and windowed H_0 features were weak, often degenerate, and did not outperform simple controls. A separate fixed-decoding GSM8K branch yielded a stronger operational result: non-convergence, as measured by token-cap termination, was the dominant predictor of error.

The main positive result emerged only after changing the task. We introduced a procedural micro-world benchmark with nonce vocabularies, held-out templates, and exact tri-label semantics over `True`/`False`/`Unknown`. In that setting, a consistent dissociation appeared across Qwen and Gemma families [7, 5]: decoder behavior systematically under-expressed `Unknown`, yet verdict-region hidden-state probes recovered substantial `Unknown` signal on held-out worlds. This dissociation survived prompt-path controls, constrained label decoding, base-vs-instruct checks, verdict-step logit analysis, and layer sweeps. The paper's final claim is therefore narrower and stronger than the original repository framing: semantic non-entailment is internally represented, but often under-realized by decoder readout.

1 Scope and Final Claim

This project began as a search for a global internal marker of correctness. The final evidence supports a different claim:

> In a held-out procedural semantic task, non-entailment (`Unknown`) is represented in hidden states but systematically under-realized by decoder outputs.

The paper therefore has three parts:

1. a negative replication record for global-topology correctness signals on GSM8K-style traces,
2. a fixed-decoding convergence analysis showing cap-linked non-convergence as a dominant error mode, and

3. a positive result on a procedural micro-world benchmark showing a replicated representation–decoder dissociation for non-entailment.

The main contribution is *not* a universal scalar "topology of truth." The stronger result is a readout-bottleneck interpretation supported by multiple controls. To avoid over-claiming, we explicitly do *not* claim:

- a universal geometry-of-truth detector across unrestricted tasks,
- a complete causal account of decoder behavior,
- or that topology alone is the primary predictive tool in natural reasoning traces.

The claim is intentionally task-conditioned and mechanism-narrow.

2 Why the Original Idea Failed

The original working hypothesis was that correct reasoning might have a cleaner global geometric or topological signature than incorrect reasoning [3, 2]. The basic recipe was:

1. run a small model on a reasoning task,
2. collect token-level hidden states,
3. treat those hidden states as a point cloud,
4. compute topological summaries such as connected-component fragmentation (H_0) and loop structure (H_1), and
5. ask whether those summaries predict correctness.

The hypothesis was attractive, but the experiments exposed three problems.

Length confounding. Prefix-based features changed largely because more points accumulated, not because the underlying reasoning became more truthful.

Degenerate window geometry. Fixed-size window preprocessing, especially per-window standardization and whitening, often collapsed windows into nearly geometry-equivalent objects. This made several windowed H_0 summaries almost pure functions of window size.

Task contamination. On natural-language math traces, hidden states reflect too many entangled factors at once: reasoning quality, verbosity, formatting drift, stop conditions, and non-convergence. This makes a single global topology scalar too blunt to isolate correctness.

The failure of the first branch is itself important. It prevented a misleading positive claim and forced a cleaner task design.

3 Experimental Trajectory

3.1 Why These Three Phases Belong in One Paper

The paper is not a grab-bag chronology. It is a causal progression in experimental design:

1. **Phase A falsified** the original global-scalar hypothesis under realistic trace conditions.
2. **Phase B diagnosed** the dominant confound in that setting (non-convergence under fixed decoding).
3. **Phase C redesigned** the task to isolate semantics directly, enabling a sharper representation-vs-readout test.

The positive result is only interpretable in light of that redesign logic.

3.2 Phase A: Global Topology on GSM8K (Negative)

Small paired runs on Qwen3.5-0.8B and Qwen3.5-2B produced weak dynamic-H_0 signals on non-capped 2B traces. The primary dynamic pilot used 10 paired 2B samples, with one capped trace removed for the non-capped comparison set ($n = 9$: 4 correct, 5 wrong). Representative pilot numbers were:

- `h0_entropy_final AUC = 0.55`
- `peak_h0_entropy AUC = 0.55`
- `auc_h0_entropy_per_token AUC = 0.50`
- `max_delta_h0 AUC = 0.60`

A corrected fixed-window test did worse:

- controls-only AUC: 0.60
- corrected-topology-only AUC: 0.20
- controls plus corrected-topology AUC: 0.20

These numbers are not compatible with a strong correctness-prediction claim. We treat them as descriptive pilot outcomes, not inferential proof that topology is universally uninformative.

3.3 Phase B: Fixed-Decoding GSM8K (Convergence Result)

A second branch held decoding fixed for Qwen3.5-2B and shifted focus from topology to operational failure mode on a GSM8K slice [4]. The key variables were:

$$C = \mathbf{1}[\text{run terminated by } \texttt{max_new_tokens}], \qquad Y = \mathbf{1}[\text{answer is wrong}].$$

On a within-question batch of $n = 180$ samples:

- capped: 87 samples (75 wrong / 12 correct)

- EOS: 93 samples (17 wrong / 76 correct)
- AUC($C \to Y$): 0.839
- odds ratio (wrong given cap): 26.40

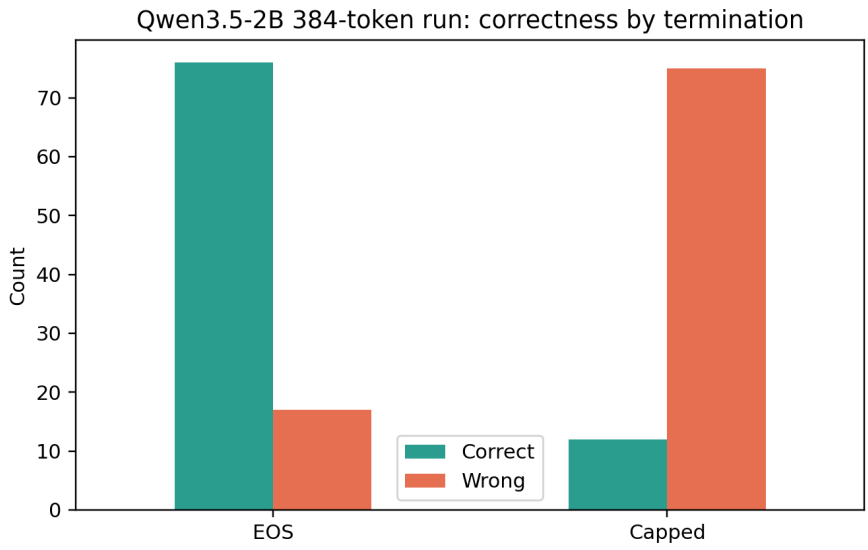

Figure 1: Fixed-config Qwen3.5-2B run: correctness split by termination mode.

The odds ratio calculation is:

$$\mathrm{OR} = \frac{\frac{75}{12}}{\frac{17}{76}} \approx 26.40.$$

A matched sensitivity rerun at 640 tokens for the originally capped cases showed that both truncation and deeper non-convergence mattered: some failures were rescued by a longer budget, but many remained wrong even after receiving more room to continue. Concretely, among 75 originally capped wrong runs, 27 flipped to correct and 48 remained wrong; 24 stayed both capped and wrong at 640. This rules out the trivial interpretation that the cap signal is only an artifact of an arbitrary token budget.

3.4 Phase C: Procedural Micro-World Semantics (Main Positive Result)

The project became scientifically cleaner only after replacing benchmark reasoning with a procedurally generated semantic task, in line with controlled-evaluation guidance from recent LM analysis literature [6]. Each generated world contains:

- a latent structured state,
- nonce entity names, attribute names, and relation names,
- held-out lexicons across train/dev/eval,
- held-out paraphrase templates across train/dev/eval,

- exact tri-label truth conditions over `True`, `False`, and `Unknown`.

Each world yields 72 examples: 9 propositions times 8 paraphrases. The main sweeps used 20 train worlds and 20 eval worlds. The generator itself produces larger split files (default 100/25/100 worlds for train/dev/eval), and analysis subsets are explicitly selected from those outputs.

Critically, anti-shortcut controls are built into generation:

- split-specific nonce lexicon pools are disjoint for entities/attributes/relations,
- eval paraphrases use template variants 4–7, while train uses 0–3 (dev uses overlap variants by design for intermediate stress),
- per-world proposition sampling is label-balanced over `True`/`False`/`Unknown`.

These details target reviewer concerns about lexical leakage and template-only clustering.

4 Task Definition

Let a world be a finite structured state

$$w = (E, A, R),$$

where E is a set of entities, A denotes unary predicates (attributes), and R denotes binary predicates (relations). A proposition ϕ is rendered into natural language by a paraphrase template. Its semantic status is one of three classes:

$$y \in \{\texttt{True}, \texttt{False}, \texttt{Unknown}\}.$$

The labels are exact:

- `True`: the world entails ϕ,
- `False`: the world entails $\neg\phi$,
- `Unknown`: neither ϕ nor $\neg\phi$ is entailed.

This matters because `Unknown` is not a stylistic hesitation label. It is exact non-entailment under the generator.

4.1 Worked Micro-World Example

Table 1 shows a compact latent world. Table 2 then shows one proposition per label class, plus a second paraphrase for one query to make template variation explicit.

Table 1: Worked world facts (illustrative).

`mep is falm.`	`grel is not falm.`	`nalo foshes sop.`

Table 2: Label semantics on the worked world.

Query statement	Gold label	Why
`The object mep has property falm.`	True	Explicit positive fact in world state.
`The object grel has property falm.`	False	Explicit negative fact (`grel is not falm`).
`The relation fosh holds from mep to nalo.`	Unknown	Neither positive nor negative fact provided for this ordered pair.
`The ordered pair (mep, nalo) has relation fosh.`	Unknown	Same proposition as previous row, different paraphrase template.

5 What Is Stored for Each Example

For each example, the pipeline stores:

- the rendered prompt,
- the gold label,
- the emitted label,
- correctness and parse status,
- token IDs,
- selected hidden states,
- verdict-step logits for canonical label tokens.

The key state locations are:

- `final_prompt`
- `prompt_tail_mean`
- `verdict_token`
- `verdict_span_mean`

These are the local decision states of the model near label emission. In code:

- `final_prompt`: last hidden vector of prompt tokens,
- `prompt_tail_mean`: mean of last $\min(5, \text{prompt_len})$ prompt vectors,
- `verdict_token`: first generated-token hidden vector (zero vector if no generated token in non-sweep extraction),
- `verdict_span_mean`: mean of first $\min(3, \text{gen_len})$ generated vectors (zero vector if none).

5.1 Inference and Prompt Protocol

Micro-world inference runs use short deterministic decoding (`temperature=0`, `max_new_tokens=4`) with strict label parsing. "No-think" in this paper refers to runs where internal reasoning mode was disabled via the `enable_thinking=False` generation flag. We evaluate both default chat-style prompting and raw/base-label prompt paths as controls.

6 Methods

6.1 Linear Probe

Take a hidden state from one example at one chosen location:

$$h \in \mathbb{R}^d.$$

Each example provides a pair (h_i, y_i), where $y_i \in \{1, 2, 3\}$ corresponds to `True`, `False`, or `Unknown`.

A linear probe is

$$W \in \mathbb{R}^{3\times d}, \qquad b \in \mathbb{R}^3$$

with class scores

$$s = Wh + b$$

and prediction

$$\hat{y} = \arg\max_c s_c.$$

Training minimizes multiclass cross-entropy:

$$\mathcal{L}(W, b) = -\sum_i \log \frac{\exp((Wh_i + b)_{y_i})}{\sum_c \exp((Wh_i + b)_c)}.$$

This probe is intentionally weak and follows the standard linear-probe setup [1]. If it succeeds, the information is already arranged in hidden space in a directly readable linear form. Implementation details (from the committed probe script):

- feature standardization: `StandardScaler(with_mean=True, with_std=True)`,
- classifier: multinomial logistic regression (`lbfgs`, $C = 1.0$, `max_iter`=4000),
- class labels: fixed order {`True`, `False`, `Unknown`},
- no class weighting, no hidden-state PCA, no extra feature engineering.

Train/test separation is by world split manifests (train worlds for fitting, held-out eval worlds for reporting), not random sentence-level splitting.

6.2 Within-World Geometry Gap

For one world and one state key, with distance $d(\cdot, \cdot)$:

$$D_{\text{same}} = \mathbb{E}[d(h_i, h_j) \mid y_i = y_j],$$

$$D_{\text{diff}} = \mathbb{E}[d(h_i, h_j) \mid y_i \neq y_j].$$

Define the world-level gap:

$$\Delta = D_{\text{diff}} - D_{\text{same}}.$$

A positive Δ means same-label states are more tightly organized than different-label states. The main metric is cosine distance over L_2-normalized vectors:

$$d_{\cos}(h_i, h_j) = 1 - \frac{h_i^\top h_j}{\|h_i\|_2 \|h_j\|_2}.$$

All pairwise distances are computed within world and state key before aggregation.

6.3 Sign-Test Reporting

World-level sign tests are reported as descriptive primary statistics (positive/zero/negative world counts). For calibration, one-sided exact binomial values under null $p = 0.5$ are:

- 19/19 positives: $p = 2^{-19} \approx 1.91 \times 10^{-6}$,
- 20/20 positives: $p = 2^{-20} \approx 9.54 \times 10^{-7}$.

These values are supportive but secondary to the held-out-world descriptive consistency.

6.4 Verdict-Step Label-Logit Metrics

Let verdict-step logits for canonical label tokens be

$$\ell_T,\ \ell_F,\ \ell_U.$$

Then

$$p(c \mid h) = \frac{e^{\ell_c}}{e^{\ell_T} + e^{\ell_F} + e^{\ell_U}},$$

with the Unknown competitiveness margin

$$m_U = \ell_U - \max(\ell_T, \ell_F).$$

If $m_U < 0$, Unknown is under-ranked against the strongest non-Unknown candidate. Canonical label-token scoring uses first-token variants for each label string with and without leading space. For each label, we keep unique first-token IDs and use the *maximum* first-token log-prob across those variants. This reduces tokenizer-surface artifacts from a single textual form.

6.5 Layer Sweeps

For each layer ℓ, evaluate the same probe protocol and report

$$R_U^{(\ell)} = \text{Unknown recall of the probe at layer } \ell.$$

This reveals where non-entailment is maximally linearly recoverable in the network. Sweep implementation details:

- `prompt_last`: extracted for every example and every layer,
- `verdict_token`: extracted at the first generated token; examples with no generated token are excluded from this branch via a validity mask,
- non-finite activations are replaced with zero before probe fitting (`nan_to_num`) to keep full sweeps stable.

7 Main Micro-World Results

7.1 Decoder Behavior vs Hidden-State Recoverability

Table 3 summarizes the central result on held-out eval worlds.

Table 3: Decoder behavior vs verdict-token probe on held-out eval worlds ($n = 1440$ examples per model).

Model	Decoder accuracy	Decoder Unknown recall	Probe Unknown recall
Qwen3.5-2B	0.436	0.000	0.738
Qwen3.5-4B (no-think)	0.478	0.000	0.129
Gemma-3-4B-it (no-think)	0.397	0.013	0.563

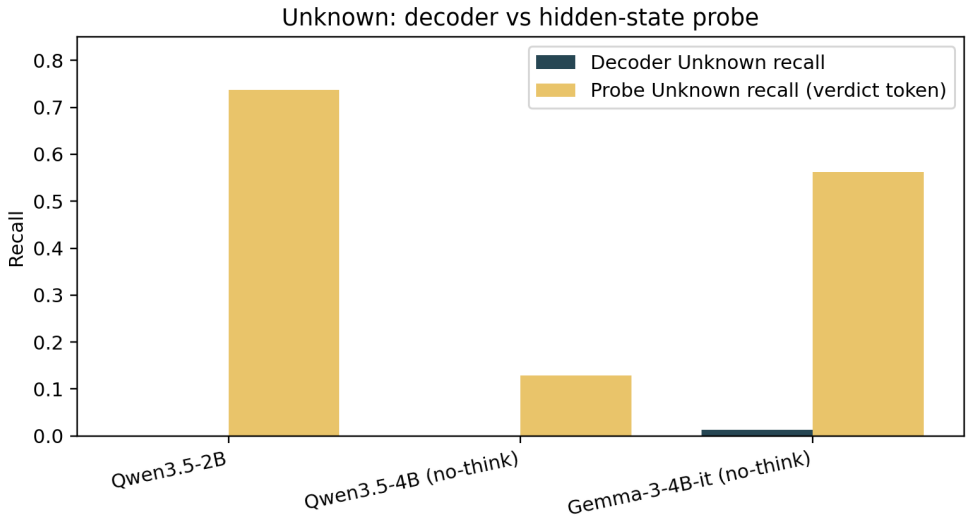

Figure 2: Unknown recall: decoder outputs vs verdict-token probe.

The key pattern is not raw accuracy. It is the gap between emitted-`Unknown` behavior and hidden-state recoverability of `Unknown`.

7.2 World-Level Geometry Consistency

World-level same-vs-different label distance gaps were positive in every evaluated world for main state keys in both Qwen3.5-2B and Gemma-3-4B-it. This includes all three class-pair comparisons (`True-False`, `True-Unknown`, `False-Unknown`) in the aggregate label-pair summaries, not only pooled different-vs-same averages. So the signal is not driven by one class boundary alone.

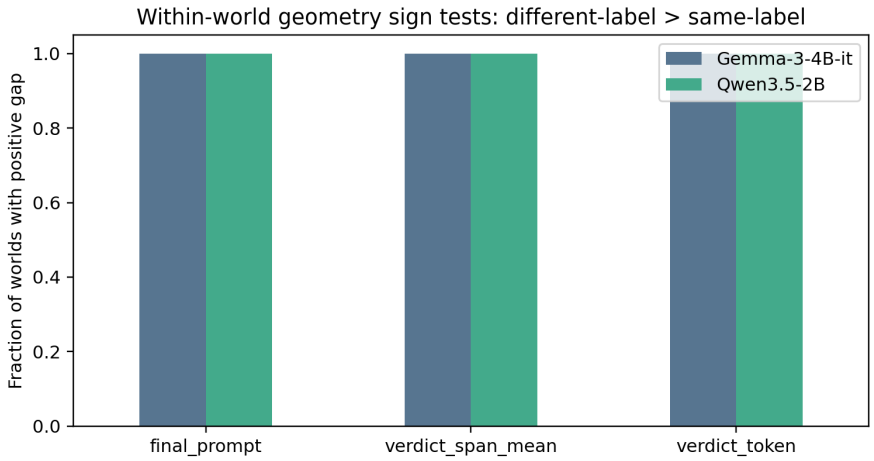

Figure 3: Fraction of worlds with positive same-vs-different label distance gap.

For the headline state keys, sign counts are 19/19 (Qwen3.5-2B) and 20/20 (Gemma-3-4B-it), matching the exact-binomial sanity values in the methods section.

7.3 Cross-Family Replication

The dissociation replicates across Qwen and Gemma:

1. decoder outputs under-express `Unknown`,
2. hidden states still encode `Unknown`,
3. the gap survives held-out worlds, held-out lexicons, and held-out templates.

8 Mechanistic Controls

8.1 Constrained Decoding

To test whether free-form decoding alone caused the issue, decoding was constrained to {`True`, `False`, `Unknown`}. This did not repair Unknown collapse in the main models:

- Qwen3.5-2B and Qwen3.5-4B remained at decoder Unknown recall = 0.0,
- Gemma-3-4B-it remained low (Unknown recall = 0.0125, unchanged in matched constrained/unconstrained eval runs).

So the bottleneck is not reducible to unconstrained text drift.

8.2 Prompt-Path Control

Raw-prompt controls were used to rule out chat-template-only explanations. The core dissociation remained. For Gemma instruct under raw prompting, decoder Unknown recall increased relative to the no-think default path, but remained materially below hidden-state recoverability, preserving the central mismatch.

8.3 Base vs Instruct

Gemma base initially showed severe parse failures under an unsuitable prompt path. After repairing prompt format with a base-specific label format, the parse confound disappeared. Yet base still showed decoder Unknown collapse while probes recovered substantial Unknown signal. Numerically:

- raw base prompt path parse-failure rate was 97.5%,
- repaired base-format path reduced parse failure to 0%,
- repaired base-format decoder Unknown recall remained 0.0.

This isolates readout behavior from trivial formatting failures.

9 Verdict-Step Label-Logit Analysis

For gold-`Unknown` decoder failures:

- Gemma-3-4B-it: mean $P(\texttt{Unknown}) = 0.192$, mean margin $m_U = -1.293$,
- Gemma-3-4B-pt (basefmt): mean $P(\texttt{Unknown}) = 0.177$, mean margin $m_U = -1.045$.

So Unknown is often present but not competitive enough at final label-token competition. Importantly, this is a *readout-stage* diagnosis: Unknown is not absent from representation, but is systematically under-ranked at the verdict step on failures where gold is Unknown. That distinction is what links probe recoverability and emitted-label collapse.

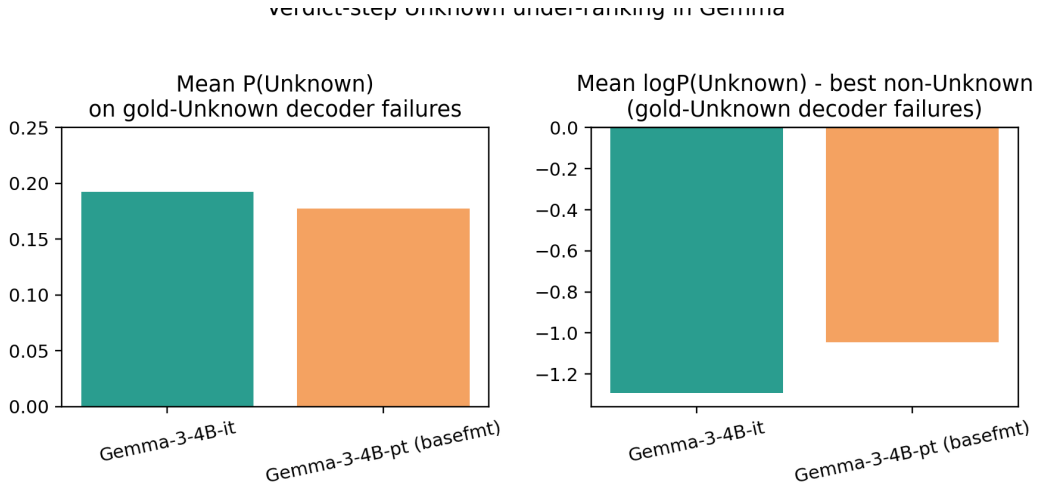

Figure 4: Unknown under-ranking at verdict step on gold-Unknown decoder failures.

10 Layer Sweeps

The best Unknown recall by state family was:

- Gemma-3-4B-it: `prompt_last` 0.779 at layer 1, `verdict_token` 0.742 at layer 10
- Gemma-3-4B-pt (basefmt): `prompt_last` 0.825 at layer 29, `verdict_token` 0.733 at layer 28

This shows strong recoverable Unknown signal exists internally in both instruct and base variants, even when emitted behavior still collapses that class. The layer locations differ by model variant (early-mid for instruct vs late for basefmt in this slice), which supports a "signal location and readout alignment" view rather than a simple "more scale always better" view.

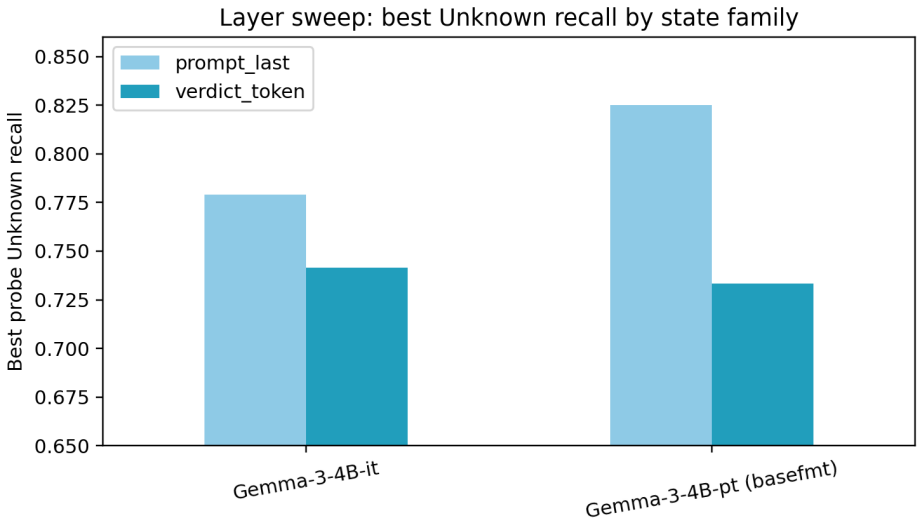

Figure 5: Best Unknown recall from layer sweeps by state family.

11 Interpretation

The combined evidence supports a readout-bottleneck interpretation:

1. semantic non-entailment is encoded internally,
2. decoder outputs under-express that class,
3. the mismatch is replicated across model families and controls,
4. verdict-step logits under-rank Unknown,
5. layer sweeps show strong internal signal upstream of failed readout.

This is stronger and narrower than the original repository framing. Results do *not* support a universal "truth topology" scalar.

12 Why This Should Survive Review

The result survives the obvious reviewer attacks:

- **Memorization objection**: nonce vocabularies, held-out lexicons, held-out templates.
- **Single-family objection**: replicated in both Qwen and Gemma.
- **Template objection**: raw-prompt controls.
- **Parse objection**: Gemma base rerun with repaired prompt format.
- **Free-decoding objection**: constrained decoding tested.
- **"No internal Unknown" objection**: probes, geometry, logits, and layer sweeps all counter it.

- **Synthetic-task objection**: synthetic design is deliberate to obtain exact entailment labels and control lexical leakage; the claim is scoped to this setting.
- **Tokenization-artifact objection**: label-logit analysis uses multiple first-token variants (with/without leading space) and takes the strongest per-label candidate.
- **Lexical-clustering objection**: eval uses held-out lexical pools and template variants, and geometry is evaluated within world across paraphrases.
- **"Probes are not causal" objection**: agreed; probe results establish information availability, while constrained decoding and verdict logits target the usage/readout side.

  No single analysis carries the paper; strength comes from triangulation.

13 Practical Artifact Map

Key outputs used in the paper:

- convergence branch: `artifacts/qwen35_2b_within_question/analysis/`
- cap sensitivity: `artifacts/qwen35_2b_cap_sensitivity_640/analysis/`
- cross-model decoder comparisons: `artifacts/micro_world_v1/comparison_decoder_*.csv`
- probe comparisons: `artifacts/micro_world_v1/comparison_probe_states_*.csv`
- constrained-decoding comparisons: `artifacts/micro_world_v1/comparison_decoder_constrained_*.csv`
- verdict-step logits: `artifacts/micro_world_v1/label_logits_*/`
- layer sweeps: `artifacts/micro_world_v1/layer_sweep_*/`

14 Implications Beyond This Benchmark

The main result has implications beyond this specific micro-world generator. If semantic non-entailment is recoverable internally while decoder outputs collapse it, then evaluation based only on emitted labels can underestimate a model's internal uncertainty structure. That matters for:

- abstention and selective prediction design,
- post-hoc confidence calibration,
- safety analysis of over-assertive outputs,
- readout-head or decoding-policy interventions that target decision alignment rather than representation learning.

This paper does not claim direct transfer to all tasks, but it motivates testing representation–readout gaps in other controlled domains.

15 Limitations

- The micro-world benchmark is synthetic, even though controlled and compositional.
- Model coverage is small-to-mid scale; larger-scale behavior is open.
- Probe recoverability does not prove causal use by decoder policy.
- The exact source of readout failure is not fully decomposed; it may involve calibration, label competition, or post-training behavior shaping.

16 Reproducibility

Locked dependency snapshot: `repro/requirements.lock.txt`

Figure generation:

```
python3 paper/scripts/make_figures.py
```

Compile:

```
cd paper
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

17 Conclusion

The original "truth topology" objective was too broad for the observed evidence. The final validated contribution is a replicated representation–decoder dissociation: semantic non-entailment is strongly encoded in hidden states but systematically under-surfaced at output time. The correct framing is no longer a search for a single topology-of-truth scalar, but a study of how semantic uncertainty is preserved internally and lost at decoder readout.

References

References

[1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. In *ICLR Workshop Track*, 2016.

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A Appendix A: Full Reproduction Commands

```
# Phase A dynamic topology
python3 scripts/analyze_paired_dynamic.py \
  --artifact-08b artifacts/paired_qwen35_08b_10 \
  --artifact-2b artifacts/paired_qwen35_2b_10 \
  --out-dir artifacts/paired_dynamic_qwen35

# Phase A corrected window test
python3 scripts/analyze_2b_window_corrected.py \
  --artifact-dir artifacts/paired_qwen35_2b_10 \
  --out-dir artifacts/qwen35_2b_window_corrected

# Phase B fixed-decoding run
python3 scripts/run_multi_sample_questions.py \
  --model-id Qwen/Qwen3.5-2B \
  --question-start 0 --question-count 20 --samples-per-question 9 \
  --temperature 0.7 --top-p 0.95 --max-new-tokens 384 \
  --artifact-dir artifacts/qwen35_2b_within_question

# Phase B cap sensitivity
python3 scripts/run_cap_sensitivity.py \
  --model-id Qwen/Qwen3.5-2B \
  --source-artifact-dir artifacts/qwen35_2b_within_question \
  --out-dir artifacts/qwen35_2b_cap_sensitivity_640 \
  --temperature 0.7 --top-p 0.95 --max-new-tokens 640

# Phase C dataset
```

```
python3 scripts/generate_micro_world_dataset.py \
  --out-dir artifacts/micro_world_v1/dataset \
  --seed 1729 --train-worlds 100 --dev-worlds 25 --eval-worlds 100 \
  --props-per-world 9 --paraphrases-per-prop 8
```

B Appendix B: Protocol Details

B.1 Dataset Generation Protocol

The generator samples partial worlds with 4–6 entities, 2 attributes, and 2 relations per world by default. Per relation/attribute assignment, facts are sampled as explicit positive, explicit negative, or omitted (Unknown) states. For each world, proposition sampling is quota-balanced over `True`/`False`/`Unknown` labels before paraphrase rendering.

B.2 Train/Test Separation

Probe training uses only train-world manifests (`status=ok`, label in {`True`,`False`,`Unknown`}). Evaluation uses held-out eval-world manifests. No sentence-level random split is used in the reported probe tables.

B.3 Probe Fitting Defaults

All reported linear probes use:

- `StandardScaler` (mean/std normalization),
- `LogisticRegression(solver=lbfgs, C=1.0, max_iter=4000)`,
- three-class label set in fixed order {`True`, `False`, `Unknown`},
- zero-division-safe precision/recall/F1 reporting.

B.4 Layer Sweep Inclusion Rules

For `verdict_token` sweeps, examples with no generated token are excluded via a validity mask. For `prompt_last` sweeps, all `status=ok` examples are included. Non-finite activations are replaced with zero prior to fitting.

C Appendix C: Definitions and Notation

- **Unknown (semantic class):** non-entailment in the generator's three-valued semantics.
- **Unknown (decoder output):** emitted label string parsed from model output.
- **Unknown recoverability:** recall of the Unknown class under a linear probe on hidden states.

- **Representation–decoder gap:** probe Unknown recall minus decoder Unknown recall on matched eval sets.

These are related but non-identical quantities, and they are reported separately throughout.